Anandakrishnan Suresh

anandakrishnansuresh003@gmail.com | linkedin.com/in/ananda-krishnan-suresh

Education

Imperial College London — London, UK
MSc Mathematics & Finance — *Sep. 2025 – Sep. 2026*

- Grade: Distinction (Expected)
- Relevant Modules: Stochastic Calculus, Statistical Methods, Deep Learning, Reinforcement Learning, Quantitative Risk Management, Simulation Methods, Numerical Methods

Christ (Deemed To Be) University — Bengaluru, India
BSc Computer Science, Mathematics & Statistics — *Jun. 2021 – Apr. 2024*

- Grade: First Class with Distinction, GPA: 3.74/4
- Relevant Modules: Time Series Analysis, Linear Regression, Statistical Inference, Probability Distributions, Linear Algebra, Real Analysis, Data Structures and Algorithms

MSc Thesis

Financial News Embeddings and Short-Term Abnormal Returns — London, UK
Imperial College London, in collaboration with Zanista AI — *Jun. 2026 – Sep. 2026*

- **Supervisors:** Dr. Arman Khaledian (external) and Dr. Philipp Jettkant (internal).
- **Research question:** Does the information in LLM embeddings of financial news predict the magnitude of subsequent short-term abnormal stock returns, and how much of this predictability depends on firm-specific information?
- **Data and preprocessing:** Analysed 550K+ news articles from Jan. 2024 to Jul. 2025. Removed near-duplicates using TF-IDF cosine similarity, reducing the corpus by 42.3%.
- **Methodology:** Embedded articles in 3,072 dimensions (`text-embedding-3-large`), termed "raw news", and constructed "pure news" ("raw news" with firm characteristics ridge-regressed out) following Didisheim et al. (2026). Compared both via PCA regressions on absolute abnormal returns at 1-, 5- and 21-day horizons, using a chronological train–test split.
- **Results:** "Raw news" outperformed "pure news" at the 1- and 5-day horizons (peak out-of-sample R^2 of 15.78% vs. 6.83% at 1 day; 11.05% vs. 4.82% at 5 days), with weak predictability at 21 days. Predictive performance correlated positively with firm identifiability ($r = 0.55$ at 1 day), and adding firm identity to "pure news" recovered "raw news" performance (R^2 of 16.01% vs. 15.78% at 1 day).

Work Experience

Software Engineer — Dubai, UAE
Pasiv Financial — *Oct. 2024 – Sep. 2025*

- Optimised Python, AWS Lambda and DynamoDB backend services for a retail investing platform, reducing critical screen load times by 40% through redesigned data-access workflows.
- Designed and implemented a trade execution pipeline integrated with broker APIs, ensuring reliable order processing and portfolio management operations.
- Resolved 80+ critical defects through debugging and validation of application logic, improving the reliability and robustness of a pre-launch financial product.

Academic Projects

Extreme Value-Based Tail Risk Modelling for Financial Returns

- Implemented GARCH(1,1) volatility modelling on Tesla (TSLA) returns prior to November 2021, using likelihood estimation and residual diagnostics to model volatility clustering and identify heavy-tailed behaviour.
- Following McNeil and Frey (2000), applied Extreme Value Theory with Generalised Pareto Distribution tails to estimate 99.5% Value-at-Risk and Expected Shortfall, providing more conservative extreme-tail forecasts than Gaussian assumptions.
- Backtested GPD-based VaR forecasts using Kupiec's unconditional coverage test, which did not reject the correct coverage at 99.5% confidence level (p = 0.112).

Reinforcement Learning Based Optimal Trade Execution

- Re-implemented the Hendricks and Wilcox (2014) optimal execution framework by constructing a custom MDP simulator using AAPL Level-2 order book data and benchmarking SARSA and DDQN against the Almgren-Chriss execution schedule.
- Outperformed Almgren-Chriss by up to 0.82 bps using hour-conditioned tabular policies and reduced downside execution risk by up to 4.8 bps using deep RL.

Additional Information

Technical Skills: Python (NumPy, Pandas, scikit-learn), AWS (Lambda, DynamoDB), SQL
Interests: Filmmaking (short films on YouTube; selected for film festival screenings)